

UTKARSH SHARMA

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Research Interests: Differentiable & Neural Rendering, Physically-Based Rendering, Inverse Graphics

Education

Indian Institute of Science (IISc)

Bangalore, India

M.Tech in Computer Science and Engineering | **CGPA: 9.1/10.0**

2024 – 2026

Key Coursework: Graphics & Visualization, Learning for 3D Vision and Inverse Graphics, Deep Learning for Computer Vision, Computational Topology, Computational Methods of Optimization, Machine Learning for Signal Processing, Design and Analysis of Algorithms

Chandigarh University

Mohali, India

B.E. in Computer Science | **CGPA: 8.24/10.0**

2020 – 2024

Research Experience

M.Tech Thesis Project: Radiosity-Guided Accelerated Path Tracing | *CS 299, IISc*

Jan – Apr 2026

- Jointly advised by **Prof. Vijay Natarajan** (Visualization & Graphics Lab) and **Prof. R. Venkatesh Babu** (Vision & AI Lab).
- Proposed a camera-independent path guiding framework utilizing hardware-accelerated progressive radiosity to extract top- K radiant contributors to construct incoming radiance distribution.
- Designed a closed-form von Mises-Fisher (vMF) mixture model constructed on-the-fly, mapping precomputed radiometric geometry to continuous spherical distributions.
- Implemented a fused CUDA update kernel achieving bounded $\mathcal{O}(T)$ memory scaling and delivering up to $2.4\times$ higher sample throughput than NPM and PPG baselines on Diffuse dominant scenes with comparable variance reduction.
- **Radiosity-Guided Accelerated Path Tracing – Utkarsh Sharma**, Prashant Goswami, Rishubh Parihar, Vijay Natarajan, R. Venkatesh Babu. (*Under review at JCGT, July 2026*)

Selected Research & Projects

Nabla: Differentiable Rendering Framework | *Python, PyTorch*

May – Jul 2026

- Implemented and validated four gradient estimators against finite-difference ground truth: autodiff, adjoint Path Replay & Radiative Backpropagation (PRB/RB), and boundary-aware Warp (Bangaru et al. 2020) and Reparameterization (Loubet et al. 2019) sampling for discontinuous visibility.
- Benchmarked PRB/RB against standard autodiff, showing up to $9.2\times$ lower VRAM usage (509 MB vs. 4,668 MB at 512×512 resolution).

Futaba: Hardware-Accelerated GPU Path Tracer | *C++17, CUDA, OptiX, OpenGL*

Mar – Aug 2025

- Built a GPU-accelerated path tracer in C++/CUDA using NVIDIA OptiX hardware ray tracing (67–145 FPS on a 382K-triangle scene), with a software BVH fallback.
- Implemented Next Event Estimation with MIS, Russian Roulette, Practical Path Guiding (Müller et al. 2017), thin-lens depth of field, environment-map importance sampling, and Beckmann microfacet BSDFs (conductor/dielectric/plastic).
- Built an interactive OpenGL viewport with multi-channel G-buffer visualization (albedo, normals, depth, heatmap) and OptiX AI denoising.

Neural Path Guiding for Path Tracing | *Mitsuba 3, PyTorch, Tiny-CUDA-NN*

Jun – Dec 2025

- Built a neural path-guiding framework in Mitsuba 3, implementing vMF mixture models, Neural Importance Sampling, and Distribution Factorization, reducing variance $2\text{--}4\times$ in complex scenes.
- Built a PyQt GUI to inspect Mitsuba renders and debug network predictions during rendering.
- Prototyped Neural Radiance Caching using online optimization and spatial hash-grid encodings.

Deep Learning-Based Denoising for Monte Carlo Rendering | *PyTorch, PBRT*

Jan – Apr 2025

- Benchmarked U-Net denoisers for low-sample Monte Carlo path tracing outputs in PyTorch and PBRT.
- Used G-Buffer features (albedo, normals, depth etc.) as multi-channel inputs to preserve geometric edges and material boundaries.
- Evaluated reconstruction quality using SSIM and PSNR on PBRT-generated test scenes.

Neural Radiance Fields & 3D Gaussian Splatting | *PyTorch*

Jan – May 2025

- Built a NeRF pipeline in PyTorch implementing the volumetric rendering equation, positional encoding, and hierarchical stratified sampling.
- Implemented 3D Gaussian Splatting, optimizing covariance matrices and spherical harmonics coefficients; validated on the NeRF Synthetic dataset (PSNR > 25 dB).

Technical Writing

Introduction to Differentiable Rendering	July 2026
– Survey covering 6 seminal papers (Li 2018 through Zhang 2023), with step-by-step mathematical derivations, PyTorch implementations of Radiative and Path Replay Backpropagation.	
Introduction to 3D Rendering	Apr 2026
– Interactive graphics guide covering the rasterization pipeline, ray tracing and SDF sphere tracing implementations.	
Importance Sampling & Path Guiding for Path Tracing	Dec 2025
– Tutorial and survey spanning 9 papers (Vorba 2014 through Dahm 2025), covering Monte Carlo integration, MIS, normalizing flows, vMF mixtures, and neural path guiding.	
3D Reconstruction with 3DGS, NeRF, and Their Variants	Nov 2025
– Survey of 12 methods from NeRF to DUST3R/InstantSplat, with derivations of volume rendering, EWA splatting, hash encoding, and sparse-view reconstruction.	

Work Experience

Netradyne	Bangalore, India
<i>Machine Learning Engineer</i>	<i>July 2026 – Present</i>
– Analyze large-scale telematics and video-derived data within the Data Science team to refine the GreenZone[®] Score , an AI-driven driver-performance metric.	
– Evaluate correlations between driver behavior indicators (e.g., proactive safe driving, risk events) and safety outcomes to improve predictive risk modeling and fleet optimization algorithms.	

Skills & Languages

Programming Languages: Python, C, C++ (17/20), CUDA, SQL, GLSL	
Graphics & 3D Vision: Physically Based Rendering (PBR), Monte Carlo Path Tracing, Differentiable Rendering, Neural Rendering (NeRF, 3DGS), Volume Rendering, NVIDIA OptiX, OpenGL	
Machine Learning & AI: PyTorch, Diffusion Models, U-Net, Neural Radiance Caching	
Libraries & Tools: Mitsuba 3, PBRT, Tiny-CUDA-NN, Dr.Jit, Pandas, NumPy, Git	
Spoken Languages: Hindi (Native), English (Fluent)	

Teaching Experience

Teaching Assistant – Algorithms & Programming <i>Indian Institute of Science</i>	Aug – Dec 2025
Teaching Assistant – Computational Topology <i>Indian Institute of Science</i>	Jan – May 2026

Academic Achievements

Rank 8 out of 123,967 in GATE Computer Science (CS) 2024 (<i>top 0.01%</i>)
Rank 17 out of 39,210 in GATE Data Science & AI (DA) 2024 (<i>top 0.05%</i>)
Winner , CodeSmash 2.0 Competitive Programming Contest
Finalist , CodeRush 3.0 Competitive Programming Contest